From the Gamma Integral to Evans–Gannon

LEAN 4 + MATHLIB

From the hyperbolic gamma construction to the Evans–Gannon equations for every odd n.

01 GAMMA FOUNDATION

INTEGRAL · CONTINUATION · TRANSFORM IDENTITIES

CONSTRUCTION · REGULARIZED INTEGRAL
`GammaContinuation.gamma`
BSS integral · finite-product continuation

CORE · SHIFTS, POLES & GROWTH
`GammaContinuation.core`
Reflection · residues · uniform ratio bounds

TWO-GAMMA · FOURIER EVALUATION
`two_gamma_explicit`
Dual shifts · sech normalization · crossed poles

Hyperbolic gamma and its identities

An explicit integral and contour transformations.

Each box groups related Lean results.

FOUR-GAMMA · CORRECTED CONTOURS
`K4_line_explicit · K4_four_term`
Residues · line independence · evenness · swap

contour tools

EULER · PARAMETER CONTINUATION
`K4_euler`
Fourier insertion · Fubini · moving pole clusters

FULL INTERFACE · SPECIALIZED PERIODS
`ruijsenaarsHypGamma`
Analytic input for the coefficient construction

02 FINITE FOURIER ANALYSIS

Gamma input throughout this stage

Sampling, contour bounds, and transform identities.

SAMPLING · CONCRETE COEFFICIENTS
`Concrete.toCoeffData`
$F(z) = 1/\gamma(\omega_2 + z)$ · explicit odd-order matrix

PRODUCTS · CONTOUR SAMPLING
`left_product · right_product`
Product transforms + negative-column transform

COLUMN FOURIER · QUADRATICS
`column_fourier · Q_eq_zero`
All quadratic residuals vanish

TWO INPUTS TO COMPLETION
Q = 0 + restricted Fourier vanishing

RESTRICTED · FOURIER VANISHING
`restricted_fourier`
Euler + four-term identity cancel the bracket

03 ALGEBRAIC COMPLETION

COMPLETION · ALL CUBICS
`CoeffData.C_all`
Boundary cases · interpolation · induction

KERNEL-CHECKED FOUNDATIONS
propext · Classical.choice · Quot.sound
No sorrys or added axioms

COMPLETION · EVANS–GANNON
`EGEquations_concrete`
Linear, quadratic, cubic and quartic identities

MAIN THEOREM · EVERY ODD n
`exists_EGEquations_odd`
Explicit complex coefficient data · unconditional

Grouped proof stages; selected dependency arrows.

EG equations proved. Categorical reconstruction is outside this formalization.